Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

Language models can give wrong answers with high confidence. But does that mean they lack information about their mistake, or simply fail to express it? We test this by reading the model's internal state directly. Across four open models and four reasoning domains (MMLU-Pro, MATH, GPQA-Diamond, and a formally graded geometry task), a simple linear probe predicts whether an answer is correct better than the model's stated confidence in 12 of 16 model–domain settings (one tie; mean +0.09 AUROC). The gap is largest on MATH (+0.20), where many incorrect solutions remain fluent and well formed.

The signal appears to reflect the model's particular attempt, rather than just question difficulty. When we compare multiple attempts at the same question, the model's own P(True) falls toward chance while the internal probe remains predictive. The signal also transfers across domains, suggesting that different tasks share some internal representation of correctness.

Finally, we test whether this signal merely correlates with confidence or actually influences it. On Mistral-Small-24B, removing the learned correctness direction after the answer has been produced reduces the ability of stated confidence to distinguish correct from incorrect answers ($0.84 \rightarrow 0.56$ AUROC), while removing a matched random direction has almost no effect. Amplifying the same direction lowers confidence on incorrect answers and improves calibration. Together, these results suggest that language models can contain more information about the correctness of their own answers than they express in verbal confidence, and that some of this internal information is causally connected to what they report.

1 Introduction

A mathematical agent that errs is unavoidable. One that errs *and knows it* can abstain, retry, or escalate. Stated confidence is a weak guide to correctness, and weakest where mathematics is hardest. A wrong multi-step derivation still ends in a cleanly formatted final answer, and nothing on the page gives it away. Section A walks one such attempt end to end: a single wrong subtraction inside an otherwise correct derivation, after which the model *raises* its stated confidence.

We split the problem into two questions. Do models internally represent that an output is wrong (**knowing**)? Does that representation reach what they report (**saying**)? If a gap exists, it is both an unused reliability signal and a concrete claim about machine introspection.

Prior work shows that output probabilities are often better calibrated than words [4], and that the truth of *given statements* is linearly readable from activations [1, 2, 6]. Verbalized-

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calibration studies measure what models say [5, 7]. We combine these threads on the model's *own attempts*. We elicit confidence before and after each attempt, measure blind (no-feedback) updating, take four readouts of the same latent quantity on identical records, and intervene on the representation [8]. "The model knows" is often used for five different claims at once, and our experiments separate them (Section 2). *Representation*: correctness is linearly decodable from internal state. *Specificity*: what is decoded tracks this attempt, not the difficulty of the question. *Generality*: one direction reads across domains. *Causal access*: manipulating it changes the confidence the model reports. *Expression*: stated confidence exposes only part of it, by an amount that depends on architecture. Decoding alone establishes only the first, which is why we test the other four; *knowing* is our reading of the five together.

2 Knowing versus saying

Models and domains. We test four open models with different architectures: Qwen3.6-27B (hybrid gated linear/full attention), GLM-4.7-Flash (MoE), Mistral-Small-24B (dense), and Gemma-4-26B-A4B (MoE, VLM-derived). Each runs on four domains, about 750 records per cell: MMLU-Pro, MATH, and GPQA-Diamond with programmatic grading, and a geometry-construction task (GeoGenBench) in which the model writes a construction from a natural-language request and a symbolic compiler decides each clause of the request exactly. The compiler matters because metacognition experiments live or die by their correctness labels, and it grades exactly against the formal specification: a pre-registered human study (N=200, two trained raters) found its disagreements with humans ran in one direction only, lenient and never harsh, so a `fail` label is trustworthy. Geometry is also the one generative domain; the others are answer-selection or short-answer tasks. We dropped benchmarks the models have saturated, since self-knowledge of failure cannot be studied where there are no failures.

Protocol. Each problem takes three turns: prospective confidence, the attempt, and retrospective confidence. Grading is external and never shown to the model, so any post-attempt revision is blind self-assessment. We take four readouts of the same latent quantity on identical records: stated confidence (0–100); P(True), the model's probability of answering "True" when asked whether its answer is correct; the answer's log-probability; and a linear *probe*. The residual stream is the model's internal state, a vector of a few thousand numbers at every token and layer. The probe is a logistic regression trained to read correctness from that vector at the confidence decision token, the position that is about to produce the confidence digit. It is scored out-of-fold with question-grouped folds and reported at one pre-specified layer at 0.7 of network depth rather than the best layer. Our metric throughout is AUROC, the probability that a random correct attempt scores above a random wrong one: 0.5 is chance, 1.0 is perfect. Four controls guard the readout. The probe adds +0.22 mean AUROC over a surface-features baseline (answer length and shape) in 15 of 16 cells. The read token is identical across records, so layer-0 activations carry no signal by construction (per-fold AUROC 0.500); anything decoded deeper was computed during the attempt. A within-question control holds difficulty fixed. A deviations log records four bugs that planned controls caught; the affected control analyses were fixed and rerun. The four Mistral cells recaptured after these fixes reproduce the headline but give a less uniform surface margin (+0.21 on MMLU-Pro, +0.07 on MATH, −0.11 on geometry, where a failed construction is visibly malformed).

Representation: knowing exceeds saying, most where errors look clean. The probe beats stated confidence in **12 of 16** cells (Table 2, left). The three losses are small-n cells, two of them Gemma-4. The gap is widest on MATH (mean +0.20). Stated confidence there is weak, while the internal signal is the strongest of any domain (0.95 on Qwen3.6, 0.83 on the recaptured Mistral cell). On MATH the spoken channel does not merely fail but moves the wrong way: with the grade never revealed, Mistral *raises* its stated confidence after attempting, and more on the attempts it got wrong (+22) than on those it got right (+16), while the probe on the same records reads 0.83. Elsewhere blind revision runs the expected way (geometry, MMLU-Pro) and is null on GPQA.

Table 1: Mistral, fresh 2,614-record capture. Left: cross-domain probe transfer (rows train, columns test; diagonal in-domain out-of-fold; Geo = GeoGenBench). Right: intervening on the model's own correctness direction at the confidence token, gain 1 an exact no-op, against a random direction rescaled to the same per-record perturbation magnitude. Removing the direction (gain 0) leaves stated confidence unable to separate successes from failures; removing the random one does not.

train ↓ test →	**Geo**	**MMLU-Pro**	**MATH**	**GPQA**
Geo	**0.68**	0.67	0.78	0.56
MMLU-Pro	0.70	**0.75**	0.84	0.59
MATH	0.80	0.72	**0.83**	0.57
GPQA	0.60	0.69	0.74	**0.57**

gain	**0**	**1**	**2**	**4**
steer, fail / ok	**97 / 98**	84 / 96	70 / 94	**54 / 88**
steer, stated AUROC	**0.56**	0.84	0.85	0.82
random, stated AUROC	0.83	0.84	0.84	unparseable

Specificity: the signal tracks the attempt, not the question. A skeptic's alternative is that the probe has only learned which questions are hard. A first control holds the question fixed. Within a single question, across its five sampled attempts, only attempt-level information can discriminate; in representative cells the model's own P(True) falls toward chance (0.47–0.59) while the probe holds (0.72–0.74). P(True) largely recovers *which questions are hard*. The probe tracks *whether this attempt succeeded*.

A sharper version tests directions rather than scores (Table 2, right). The pre-attempt read sits at the last prompt token; within a question it is identical across the five attempts, so it can encode difficulty but not the outcome of a sampled attempt. Train the post-attempt readout, then evaluate the same fitted pipeline at that pre-attempt site. On Mistral MMLU-Pro it is chance before the attempt (0.490) but strong after it (0.754), and residualizing away the pre-attempt score leaves 0.751. This is the cleanest evidence that the direction encodes *this attempt went wrong*, not just *this question is hard*. MATH is mixed, with both difficulty and attempt-level signal. GPQA is weak throughout. Geometry is the boundary case: the pre-site already predicts failure and the post readout transfers back before the attempt, so this cell is mostly difficulty or template structure rather than attempt-local self-assessment.

Generality: a substantially shared correctness direction. If the model had one internal sense of "this is wrong," a probe trained on one subject should work on another. It does: a probe trained in one domain and evaluated in another keeps most of its in-domain AUROC. In the original matrix, retention was about 87–90% (Qwen3.6 off-diagonal 0.71 vs. diagonal 0.81), and sometimes lossless within QA (Qwen3.6 MMLU-Pro→MATH 0.86 vs. 0.91). A fresh Mistral capture across all four domains, analysed with a corrected standardization step, gives 97% retention (Table 1: off-diagonal mean 0.69 vs. diagonal 0.71), with direction cosines of 0.34–0.82. Transfer into GPQA is bounded by its weak in-domain signal (0.57). On Mistral the geometry anomaly does not recur: a MATH-trained probe reads geometry at 0.80, above geometry's own in-domain 0.68, so the anti-transfer below is specific to Gemma-4.

Geometry is partly special on one model. On Gemma-4, geometry-trained probes *anti-transfer* to QA (0.31–0.39), while QA-trained probes work on geometry (about 0.70). Probes trained on the generative task can latch onto task-specific features; QA-trained probes find the general direction. This is a shared correctness subspace rather than a single invariant axis, which is what a deployable readout needs and what a workspace account (below) predicts.

Causal access: stated confidence depends on the direction. Reading shows the signal exists; intervening shows the report depends on it. We intervene at the confidence token, after the answer is written, so the answer cannot change, scaling each record's own projection onto the correctness direction by a gain g (g=1 is an exact no-op). The direction is supervised: learned from labeled examples, applied without per-example labels at intervention time. Numbers are Mistral × MATH, 150 held-out records, bootstrapped over questions, against a random direction rescaled to the same perturbation magnitude. Mean-ablating the direction (g=0) drops the AUROC of *stated* confidence from 0.836 to 0.564, -0.270 $[-0.422, -0.122]$; ablating the matched random direction does nothing, -0.002 $[-0.032, +0.025]$. This does not look like generic damage: parse rate stays at 1.00 and confidence on wrong answers *rises* from 84 to 97, converging with correct answers at 98. Amplifying instead drives failed-answer confidence to 70 and 54 at g=2, 4 while correct answers hold near 90, halving calibration error, 0.34 → 0.19 (Table 1, right); dampening increases overconfidence. Amplification does not reliably improve *discrimination*, so the direction carries the information that makes stated confidence informative at all and the gain sets how much of it reaches the number. Two controls place it: the same intervention at 0.3 depth does nothing ($+0.005$ $[-0.002, +0.015]$), reaching full effect by 0.7; and a direction fitted on MMLU-Pro, where a surface baseline of 0.54 against a probe of 0.75 rules out "my output looks malformed", still governs MATH's self-report (0.880 → 0.709).

Expression: a label-free witness, with architectural limits. Trained probes invite an overfitting critique, so we added an unsupervised instrument: the Jacobian lens of the global-workspace account [3]. The lens is fit on generic text, with no task labels anywhere in its construction, and scores the same stored activations as a wrong-vs-correct disposition. On dense models it lands in probe territory (Mistral 0.76–0.82; Qwen2.5-14B, a dense within-family control, 0.88) and beats stated confidence with no labels. Re-scoring those readouts against the recaptured Mistral cells reproduces it without labels: 0.82 on MATH and 0.75 on MMLU-Pro, against a supervised probe at 0.83 and 0.75, layer-0 control 0.49–0.51. Two independent instruments find one direction. Because the lens reads through the model's own output pathway, the signal it finds is *positioned to influence that pathway*; whether it is verbalizable as a concept is a further question we do not test. On the other architectures the lens itself fails (linear-attention hybrid 0.43, MoE 0.31) while the supervised probe still reads the signal in every model. Transported through the averaged map, the lens's "correct" and "wrong" readout vectors collapse to cosine 0.96 on MoE, against about 0.4 on dense models. An averaged map is only faithful when the routing barely varies per input, and both failing architectures vary theirs: a mixture of experts by which expert fires, and a gated linear-attention stack by its input-dependent gates. Swapping that hybrid for a dense model of the same family flips the lens from 0.43 to 0.88. Our reading is narrower than the mechanism: the label-free shortcut depends on architecture, while supervised probes still find the signal. A direct per-input wiring-variance measurement is the next test; without it, the workspace account remains a hypothesis for the expression gap. The question is no longer only *whether* models carry the signal, but *what gates the knowing from the saying.*

3 Discussion

For reliable mathematical agents. Where a domain admits an exact verifier, as geometry does, use it. Where it does not, the internal signal is the best attempt-specific readout we found: abstaining on the probe's lowest-scoring outputs raises accuracy far more than abstaining on low stated confidence, and still works in cells where verbalized abstention fails outright. And because the report depends causally on the direction, stated confidence can be made honest by amplification rather than retraining.

Limitations. An answer-end read, taken at the last token the model writes and before any confidence question exists, finds the signal already present on MMLU-Pro (0.69, against 0.57 from the question alone), so it is generated by attempting rather than by being asked; the prompt still sharpens it to 0.77. MATH cannot separate the two, its question-only read being 0.81 already. A probe could also key on properties of failed derivations rather than a dedicated metacognitive variable; our surface baseline bounds this without closing it, more tightly on MMLU-Pro than on MATH or geometry (Section C). The causal arm is cleanest on one model, with architectural replication under way, and two MATH cells ran at reduced n. Whether the knowing–saying gap is an artifact of preference tuning is open; a base-model arm would test it.

Takeaway. The conservative claim is not that a probe proves introspection. It is that mathematical models carry attempt-specific error information that their stated confidence often fails to report, and that on Mistral the reporting channel depends on that direction. Reliability work should read, test, and route that signal rather than treating the spoken confidence number as the whole story.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B $\times$ MATH cell, summarised first and then given in full.

One record, Mistral-Small-24B $\times$ MATH. *What integer n satisfies* $0 \leq n < 18$ *and* $n \equiv -11213141 \pmod{18}$*?* The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 3$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: P(True) 0.32, mean answer log-prob -0.05.

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while P(True) on the same transcript assigns only 0.32 probability to correctness. The rest of this appendix gives the prompts and the full transcript.

A.1 Prompts

Each record is four forward passes. A read is first taken at the last prompt token of the bare question, before any elicitation, then three turns are generated. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed `Confidence: N` format is what makes the read site content-neutral: the digit token has identical local context on every record, so layer-0 activations carry no signal by construction.

System prompt (MATH).

```
Solve the problem step by step. Put your final answer in \boxed{}.
```

Turn 1, prospective confidence. The problem is followed by:

```
Before answering, estimate how likely you are to answer this question
correctly. Reply with EXACTLY one line and nothing else:
Confidence: N
where N is an integer from 0 (certain to be wrong) to 100 (certain to be
correct). Do NOT answer yet.
```

Turn 2, the attempt. The turn-1 reply is left in context as the assistant message, then:

```
Now give your final answer inside \boxed{}.
```

Turn 3, retrospective confidence. The attempt is left in context, then:

```
Now assess the answer you just gave. On a scale of 0 to 100, how confident
are you that it is correct? Reply with exactly one line and nothing else:
Confidence: N
```

P(True) baseline. A separate teacher-forced branch off turn 2, no generation. Next-token mass on `True` versus `False`:

```
Consider your final answer above. Is it correct? Reply with exactly one word:
True or False.
```

Grading is external and is never shown to the model, so the turn-3 revision is blind self-assessment.

A.2 One record, verbatim

Record `math_test_number_theory_753`, sample 2 of 5.

Problem. *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?*

Turn 1 reply. `Confidence: 90`

Turn 2 reply (complete).

> First, we need to find the remainder when -11213141 is divided by 18.
> 1. Calculate $-11213141 \bmod 18$:
>
> $$-11213141 \div 18 \approx -622952.2778$$
>
> The integer part of the division is -622953.
>
> $$-11213141 = 18 \times (-622953) + r$$
>
> where r is the remainder.
> 2. Calculate the remainder r:
>
> $$r = -11213141 - 18 \times (-622953)$$
> $$r = -11213141 + 11213154$$
> $$r = 3$$
>
> Therefore,
>
> $$-11213141 \equiv 3 \pmod{18}$$
>
> So, the integer n that satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

Turn 3 reply. `Confidence: 95`

What happened. The method is right and every intermediate is right: $18 \times 622953 = 11213154$ checks out, and the model has written the correct subtraction on the line above its answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing in the presentation marks the slip. The derivation is well-formed, correctly structured, and confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an attempt that failed.

Readouts of this attempt.

readout	**value**
stated confidence, prospective (turn 1)	90
stated confidence, retrospective (turn 3)	95
P(True)	0.32
mean answer log-probability	−0.05
graded outcome	**incorrect**

This is one record, chosen to illustrate the failure mode rather than to carry evidential weight; the cell-level numbers are in Table 2.

B Calibration beyond ECE, and cell-level multiplicity

ECE depends on binning and on sample composition, so we report two binning-free proper scoring rules alongside it. On Mistral × MATH (750 records, accuracy 0.648), the model's retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence, worse calibrated in absolute terms. The steering results in Table 1 are reported as ECE because the steered runs stored per-gain aggregates rather than per-record confidences; recomputing Brier and NLL at each gain requires re-running the intervention and is the first thing we would add.

The cell-level bootstrap counts in Table 2 are uncorrected. With 16 cells, some individually significant results are expected under the null, so the cell counts should be read as descriptive. The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells with one tie, significantly positive in 10, and never significantly negative. A hierarchical model pooling across cells with model and domain as random effects is the right test and is not yet run.

C Alternative explanations for the probe signal

A probe could key on properties of failed derivations rather than on a dedicated metacognitive variable: low token probability, hedging language, or, on MATH, an early-layer channel next to the answer text. Our surface-features baseline (answer length, digit count, line count, word count, brace count) bounds this without closing it, and the bound is domain-dependent: the probe adds +0.21 over surface on MMLU-Pro but only +0.07 on MATH and −0.11 on geometry, where a construction that fails to compile is visibly malformed. Two results argue against a pure surface account on the domains where the signal is strong. A direction fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs MATH's self-report under ablation ($0.880 \rightarrow 0.709$). And the pre-attempt read, taken where the activation is byte-identical across a question's attempts, cannot recover the post-attempt direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would predict it could. A linear probe at one token also undercounts nonlinear or multi-token signal, so the reported numbers are a floor.

D Per-cell numbers and specificity controls

Table 2: Post-attempt correctness discrimination (AUROC). Left: stated confidence vs. the probe in representative cells; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Section B). Right: Mistral specificity controls. "pre" is the last prompt token before the attempt; "post" is after the attempt; "resid." removes the pre score; "post→pre" applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

model × domain	**stated**	**probe**
Gemma-4 × MATH	0.57	**0.78**
Qwen3.6 × MMLU-Pro	0.67	**0.84**
GLM × MMLU-Pro	0.67	**0.85**
Mistral × MATH	0.79	**0.83**

Mistral	**pre**	**post**	**resid.**	**post→pre**
MMLU-Pro	0.568	0.754	0.751	0.490
MATH	0.792	0.834	0.701	0.624
GPQA	0.593	0.572	0.544	0.479
GeoGen	0.770	0.676	0.505	0.648